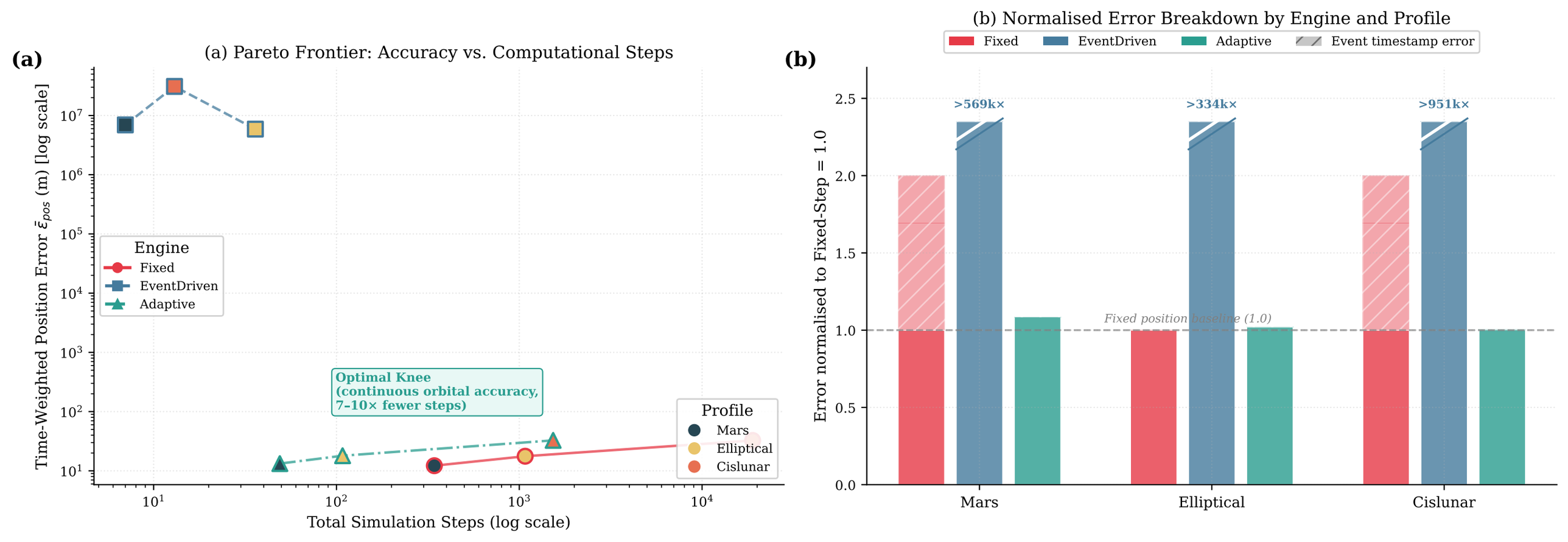
(a)
(a) Pareto Frontier: Accuracy vs. Computational Steps
Time-Weighted Position Error $\bar{\varepsilon}_{pos}$ (m) [log scale]
Total Simulation Steps (log scale)
Engine
Fixed
EventDriven
Adaptive
Optimal Knee
(continuous orbital accuracy,
7–10× fewer steps)
Profile
Mars
Elliptical
Cislunar
(b)
(b) Normalised Error Breakdown by Engine and Profile
Fixed
EventDriven
Adaptive
Event timestamp error
Error normalised to Fixed-Step = 1.0
>569k×
>334k×
>951k×
Fixed position baseline (1.0)
Mars
Elliptical
Cislunar